Machine Learning Assignment.

1. Which of the following methods do we use to find the best fit line for data in Linear Regression?

Ans. Least Square Error

2. Which of the following statement is true about outliers in linear regression?

Ans. Linear regression is sensitive to outliers

3. A line falls from left to right if a slope is _____?

Ans. Negative

4. Which of the following will have symmetric relation between dependent variable and independent
variable?

Ans. Regression

5. Which of the following is the reason for over fitting condition?

Ans. Low bias and high variance

6. If output involves label then that model is called as:

Ans. Predictive model

7. Lasso and Ridge regression techniques belong to _____?

Ans. Regularization

8. To overcome with imbalance dataset which technique can be used?

Ans. SMOTE

9. The AUC Receiver Operator Characteristic (AUCROC) curve is an evaluation metric for binary
classification problems. It uses _____ to make graph?

Ans. TPR and FPR

10. In AUC Receiver Operator Characteristic (AUCROC) curve for the better model area under the
curve should be less.

Ans. True

11. Pick the feature extraction from below:

Ans. Apply PCA to project high dimensional data

12. Which of the following is true about Normal Equation used to compute the coefficient of the Linear
Ans. Option A, B & C

13. Explain the term regularization?

Ans. Regularization in machine learning is used in regression model to prevent the overfitting and underfitting of the model.

14. Which particular algorithms are used for regularization?

Ans. LASSO, Ridge, Elastic net is used for regularization. LASSO only consider those features which are important for model so we can avoid overfitting and underfitting of model. While ridge the features are prioritized accordingly such that our model will not get overfitting and underfitting. Elastic net is the combination of LASSO & Ridge.

15. Explain the term error present in linear regression equation?

Ans. It means there is a difference between actual and predicted values of labels.

Python Worksheet1.

1. Which of the following operators is used to calculate remainder in a division?

Ans. %

2. In python 2//3 is equal to?

Ans. 0

3. In python, 6<<2 is equal to?

Ans. 24

4. In python, 6&2 will give which of the following as output?

Ans. 2

5. In python, 6|2 will give which of the following as output?

Ans. 6

6. What does the finally keyword denotes in python?

Ans. the finally block will be executed no matter if the try block raises an error or not.

7. What does raise keyword is used for in python?

Ans. It is used to raise an exception.

8. Which of the following is a common use case of yield keyword in python?

Ans. In defining a generator

9. Which of the following are the valid variable names?

Ans. Option A, B & C

10. Which of the following are the keywords in python?

Ans. Yield & Raise

Statistics Worksheet1.

1. Bernoulli random variables take (only) the values 1 and 0.

Ans. True

2. Which of the following theorem states that the distribution of averages of iid variables, properly normalized, becomes that of a standard normal as the sample size increases?

Ans. Central limit theorem

3. Which of the following is incorrect with respect to use of Poisson distribution?

Ans. Modeling bounded count data

4. Point out the correct statement.

Ans. The exponent of a normally distributed random variables follows what is called the log- normal distribution

Sums of normally distributed random variables are again normally distributed even if the variables are dependent

The square of a standard normal random variable follows what is called chi-squared distribution

5. _____ random variables are used to model rates.

Ans. Poisson

6. Usually replacing the standard error by its estimated value does change the CLT.

Ans. True

7. Which of the following testing is concerned with making decisions using data?

Ans. Hypothesis

8. Normalized data are centered at_____ and have units equal to standard deviations of the original data.

Ans. 0

9. Which of the following statement is incorrect with respect to outliers?

Ans. Outliers cannot conform to the regression relationship

10. What do you understand by the term Normal Distribution?

Ans. Normal distribution is a term for a probability distribution. Its graph is in bell curve. In mean is zero. While standard deviation is skewed right, symmetrical & skewed left

11. How do you handle missing data? What imputation techniques do you recommend?

Ans. Handling missing data is filling the empty values in data.

12. What is A/B testing?

Ans. It is used to make experiments on 2 different variants to see which one performs better on given input.

13. Is mean imputation of missing data acceptable practice?

Ans. No I don't think it is acceptable practice because it ignores the feature correlation.

14. What is linear regression in statistics?

Ans. It is an algorithm in machine learning to build a model. Its used to predict the labels based on given features. It uses best fit regression line for model building.

15. What are the various branches of statistics?

Ans. Descriptive and inferential statistics.